Summary in Graph

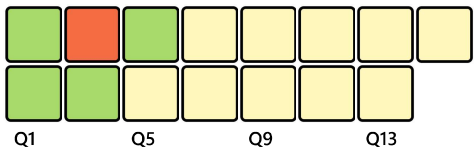
Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Topic Wise Test 1)

Qs. Attempted:	5 5 + 0	Correct Marks:	4 4 + 0
Correct Attempts:	4 4 + 0	Penalty Marks:	0.33 0.33 + 0
Incorrect Attempts:	1 1 + 0	Resultant Marks:	3.67 3.67 + 0

Total Questions:	15 10 + 5
Total Marks:	20 10 + 10
Exam Duration:	45 Minutes
Time Taken:	45 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (64) out of 637 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | AI/ML

What is the general model for $\mathbb{P}(Y = 1 \mid X = x, \theta)$ in logistic regression, where $X = (X_0, X_1, \ldots, X_n)$ is the features, Y is the predictions, and θ is the parameters? Assume that a bias term has already been appended to X (i.e., $X_0 = \mathbf{1}$).

A. $\mathbb{P}(Y = 1 \mid X = x, \theta) = \frac{1}{1+e^{-\theta^\top x}}$
B. $\mathbb{P}(Y = 1 \mid X = x, \theta) = \theta^\top x$

C. $\mathbb{P}(Y=1 \mid X=x, \theta)=\log\left(1+e^{-\theta^{\top} x}\right)$
D. $\mathbb{P}(Y=1 \mid X=x, \theta)=\log\left(1+e^{\theta^{\top} x}\right)$

Your Answer: A | Correct Answer: A | Correct | Time taken: 14min 17sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | AI/ML

Which of the following statements about logistic regression is true?

A. The loss function of logistic regression without regularization is convex, and the loss function of logistic regression with L2 regularization is convex.
B. Neither the loss function of logistic regression without regularization is convex nor the loss function of logistic regression with L2 regularization is convex.
C. The loss function of logistic regression without regularization is convex, but the loss function of logistic regression with L2 regularization is non-convex.
D. The loss function of logistic regression without regularization is non-convex, but the loss function of logistic regression with L2 regularization is convex.

Your Answer: A | Correct Answer: A | Correct | Time taken: 10min 39sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | AI/ML

In Lasso regression, what does the L1 regularization term primarily encourage?

A. Encourages the model to fit the training data more closely.
B. Encourages the model to have large coefficients for all features.
C. Encourages the model to have small but non-zero coefficients for all features.
D. Encourages sparsity by driving some feature coefficients to zero.

Your Answer: C | Correct Answer: D | Incorrect | Time taken: 00min 35sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | AI/ML

Consider a trained logistic regression model with a weight vector W and a test accuracy A on a given dataset. Assume there is no bias term in the model. If the weight vector W is divided by 2 , which of the following statements is correct?

A. The test accuracy will double.
B. The test accuracy will halve.
C. The test accuracy will remain the same.
D. The test accuracy will be unpredictable.

Your Answer: C | Correct Answer: C | Correct | Time taken: 09min 41sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | AI/ML

Suppose you have a logistic regression classifier. Consider a point that is currently classified correctly and is far away from the decision boundary. If you remove this point from the training set and re-train the classifier, which of the following statements is correct regarding the decision boundary?

A. The decision boundary will always change.
B. The decision boundary will never change.
C. The decision boundary will always remain the same.

D. The decision boundary may change, depending on the data.

Your Answer: D | Correct Answer: D | Correct | Time taken: 00min 37sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | AI/ML

Which statements are true about ridge regression and Lasso (with $\lambda > 0$ for both).

A. Ridge regression has a unique solution if and only if the design matrix has full rank.
B. There are points in feature space where the gradient of Lasso's cost function is not defined.
C. Ridge regression can be formulated as a linear programming problem.
D. One of Lasso's virtues is its tendency to set some weights to zero.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 09min 49sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | AI/ML

Suppose you train a logistic regression model with cross-entropy loss. Assume that you're updating the model using gradient descent with a sufficiently low learning rate. Which of the following are valid ways to initialize the parameters? Assume that numerical stability is not an issue (i.e. when running the model, you don't pass in an invalid argument to a function, such as 0 to $\log$).

A. Initialize all parameters with 0 's
B. Initialize all parameters with 1's
C. Randomly initialize parameters
D. All of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | AI/ML

Consider adding the elastic net regularization term $\lambda_1\|w\|_2^2 + \lambda_2\|w\|_1$ to linear regression. If we want our weights to be smaller, but dislike having many completely zero weights, what is most likely to be the best choice?

A. Increase λ_1, Decrease λ_2
B. Decrease λ_1, Increase λ_2
C. Decrease λ_1, Decrease λ_2
D. Increase λ_1, Increase λ_2

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | AI/ML

Which of the following statements are true about Lasso and ridge regression?

A. Both ridge regression and Lasso are methods used to reduce overfitting that might occur in standard linear regression.
B. The ℓ_1-norm regularization used in Lasso has a tendency to induce sparsity in the weight vector.
C. Both ridge regression and Lasso have a cost function with a minimizing weight vector w^* that we can write as a closed-form algebraic expression.

D. Ridge regression shrinks the weight vector but rarely drives its components to exactly zero.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | AI/ML

We are fitting a linear function to data in which some features are known to be much noisier than others. We account for this by applying an asymmetric penalty in ridge regression: in the normal equations, we replace the identity matrix with a different diagonal matrix, where each entry on the diagonal is a hyperparameter. Although there are many hyperparameters, suppose that we magically find the best hyperparameter values for our validation set. How is the result likely to differ from the result of standard ridge regression?

A. Lower validation error.
B. Higher training error.
C. The number of weights equal to zero tends to be greater.
D. The difference is usually small, as the hyperparameters will tend to have almost the same value.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | AI/ML

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous, smooth function whose derivative $f'(x)$ is also continuous. Suppose f has a unique global minimum $x^* \in (-\infty, \infty)$, and you are using gradient descent to find x^*. You fix some $x^{(0)} \in \mathbb{R}$ and $\epsilon > 0$, and run $x^{(t)} = x^{(t-1)} - \epsilon f'\left(x^{(t-1)}\right)$ repeatedly. Which of the following statements are true?

A. Gradient descent is sure to converge, to some value, for any step size $\epsilon > 0$
B. If f has a local minimum x' different from the global one, i.e., $x' \neq x^*$, and $x^{(t)} = x'$ for some t, gradient descent will not converge to x^*
C. Assuming gradient descent converges, it converges to x^* if and only if f is convex
D. If, additionally, f is the objective function of logistic regression, and gradient descent converges, then it converges to x^*

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | AI/ML

Although linear regression is a linear model, it can still represent non-linear functions by applying a non-linear feature transformation ϕ, which maps input data to a higher-dimensional feature space.What key assumption underlies this approach?

A. Data points that lie on a non-linear surface in the original input space are mapped to a hyperplane in the feature space.
B. Data that is linearly separable in the input space becomes linearly inseparable in a higher-dimensional feature space.
C. Data points that are near a non-linear surface in the input space are mapped close to a hyperplane in the feature space.
D. A hyperplane that minimizes prediction error in the input space corresponds directly to a non-linear surface in the feature space.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | AI/ML

We trained the logistic regression model on a set of labeled examples $\mathcal{D}$. We obtained some weight vector $\mathbf{w}$ and bias $w_0 = 0.15$. For the so-obtained model, an example $\mathbf{x}$, whose label is $y = 0$, inflicts a cross-entropy loss of $L(0, h(\mathbf{x})) = 0.274$. What would be the value of the cross-entropy loss for the example $\mathbf{x}$ if we multiplied its features by two and changed its label?

A. 2.54
B. 4.03
C. 7.11
D. 1.19

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | AI/ML

Consider using linear regression with the following training data:

$\mathbf{x}$	$\mathbf{y}$
-1	-1
0	0
1	2

Suppose you model the response as $\mathbf{y} = \theta_0 + \theta_1 \mathbf{x}$. Using least squares linear regression, what are the parameters θ_0 and θ_1 ?

A. $\theta_0 = \frac{1}{3}, \theta_1 = \frac{3}{2}$
B. $\theta_0 = \frac{1}{3}, \theta_1 = \frac{1}{2}$
C. $\theta_0 = 0, \theta_1 = 2$
D. $\theta_0 = \frac{2}{3}, \theta_1 = 1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | AI/ML

Suppose you have a dataset where the label is binary and generated by a fair coin toss and the input features are generated by sampling i.i.d. , independently of the label. Let n denote the number of examples in your dataset and d the number of input features. You perform logistic regression using a random 70/30 train-validation split. Let Acctrain denote the training accuracy and Accval the validation accuracy. Select the TRUE statements.

A. As $\mathrm{n} \rightarrow \infty$, Acctrain approaches 50%.
B. As $\mathrm{n} \rightarrow \infty$, Accval approaches 50%.
C. As $\mathrm{d} \rightarrow \infty$, Acctrain approaches 100%.
D. As $d \rightarrow \infty$, Accval approaches 100%.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

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